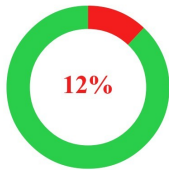


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730	5328	4.08	29.92 (88%)

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<https://www.instagram.com/reel/DNgIvV6P2WA>

This supports later review when a rule or model assumption is questioned.

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<https://www.metricstream.com/learn/what-is-contro...>

Controls need policy ownership and operational testing before use in a real environment.

#3 3% Similar

<https://delinea.com/products/secret-server/features...>

The proposed workflow checks whether the context is new, whether it is combined with unusual timing or privileged activity, and whether an approved business explanation exists.

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<https://quizlet.com/848869960/swe-chapters-8-10-f...>

Chapter 4 has documented a transparent prototype architecture, event data model, feature design, illustrative scenarios and a future test strategy.

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843	6559	2.22	34.78 (94%)

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<https://csrc.nist.rip/publications/nistpubs/800-12/80...>

An excluded or malformed record should be visible in the audit trail rather than silently changed.

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https://www.splunk.com/en_us/blog/learn/change-...

Feature values require periodic review because normal business behaviour changes with staffing, regions, maintenance windows and application configuration.

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Total Words	Total Characters	Plagiarized Sentences	Unique Sentences
371	2856	0	17 (100%)

#1 100% Unique

CHAPTER 5

RESULTS, FINDINGS, RECOMMENDATIONS, FUTURE SCOPE AND CONCLUSION

5. RESULTS, FINDINGS, RECOMMENDATIONS, FUTURE SCOPE AND CONCLUSION

The primary result of this work is a documented prototype design for AI-assisted threat hunting in a cloud ERP context. It defines how permitted audit data could be converted into contextual review signals and how an analyst could use those signals without delegating final judgement to a model.

This is not a result claim about an operating production service, measured accuracy or actual enterprise incident detection.

5.1 Findings

The analysis finds that useful threat hunting depends on the relationship between fields, not on one isolated event. Time, action rarity, failure patterns, source context, resource sensitivity and role privilege form a more meaningful review picture when considered together.

A second finding is that explainability is essential. A prioritisation score without the associated context does not provide an analyst with sufficient basis for a business-sensitive decision.

5.2 Recommendations

A future implementation should begin with a controlled, authorized pilot using sanitized or non-production data. It should define data ownership, access controls, retention rules, analyst roles, escalation paths and a small set of testable scenarios before expanding coverage.

The organisation should compare the proposed prioritisation with existing monitoring, document false positives and false negatives where labels are available, and review model and feature changes through an approved change process.

5.3 Future Scope

Future work may connect approved streaming sources, build a protected analyst dashboard, introduce role-aware baselines, evaluate multiple algorithms, capture structured analyst feedback and test the design against a labelled sandbox dataset. Any deployment should be preceded by privacy, security and operational review.

Other future directions include transparent explanation panels, broader correlation with business-process events, adversarial testing and formal evaluation of alert quality under realistic workload conditions.

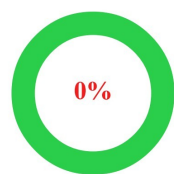
5.4 Conclusion

This report has presented an autonomous cyber-defense concept as an AI-assisted, human-governed prototype for real-time threat hunting in cloud ERP environments. The proposed design combines authorized audit telemetry, normalization, interpretable feature engineering, anomaly-oriented prioritisation and analyst triage.

The contribution is a clear, accountable foundation for future controlled implementation. It deliberately avoids unsupported claims of production deployment, verified accuracy, real screenshots or plagiarism evidence. Any future operational use must be validated with approved data, defined controls and human oversight.



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597	4486	0	24 (100%)

#1 100% Unique

CHAPTER 3

PROJECT PLANNING AND METHODOLOGY

3. PROJECT PLANNING AND METHODOLOGY

The project uses a design-and-evaluation methodology. It begins with a business-security problem, identifies permitted data sources, defines features and controls, designs a processing flow, and discusses evaluation using illustrative scenarios. This is appropriate for a prototype because the intended contribution is a structured artefact and a transparent evaluation plan rather than a claim of production results.

Each phase has a review point: scope approval, data-authorisation confirmation, feature review, scenario review and report validation. This sequencing reduces the risk of treating sensitive security data as ordinary development input.

3.1 Project Plan

The project plan separates analysis from operational deployment. Its deliverables are a documented problem statement, reviewed literature, proposed architecture, data dictionary, feature design, illustrative scenarios and a final report. This structure keeps claims aligned with available evidence.

The following schedule is a planning representation used to organise the academic work.

It is not a record of a production release.

Week Planned activity Output

1 Problem and scope definition Approved scope

2 Literature and technology review Review notes

3 Data and feature design Data dictionary

4 Architecture and workflow Design figures

5 Illustrative scenarios Analysis narrative

6 Risk and test planning Test matrix

7 Report drafting Chapter review

8 Final validation Submission-ready report

3.2 Data Governance and Collection

Data collection is limited to authorized cloud audit records, sanitized examples, publicly available cybersecurity material or synthetic scenario records. Before analysis, the design requires schema validation, field minimisation, time normalization, removal or masking of unnecessary identifiers and documentation of provenance.

The project does not claim to have ingested confidential ERP records. Where examples are presented, they are illustrative and should not be interpreted as exports from an organisation.

3.3 Feature Engineering

Feature engineering converts raw events into signals suitable for review. Examples include hour-of-day deviation, action rarity, failed-request count, resource sensitivity, role privilege, source novelty and short-window event frequency. Context is retained so that a reviewer can understand how a feature was derived.

A key design choice is to avoid opaque, single-number decision making. The score should be accompanied by the event fields and contributing conditions that prompted prioritisation.

Feature Purpose Example interpretation

Event time deviation Identify uncommon timing Privileged action outside expected window

Action rarity Highlight uncommon commands Rare administrative operation

Failure burst Capture repeated denials Repeated rejected access

Source novelty Compare with known context New region or IP context

Request rate Identify bursts Short interval API activity

3.4 Risk Management

Project risks include unapproved data use, false-positive overload, model drift, incomplete log sources, ambiguous ownership and over-reliance on an automated score. The mitigation strategy is to keep the prototype scoped, use permitted data, require analyst review, capture dispositions, document exclusions and test assumptions with controlled scenarios.

The project plan also includes communication between the student, guide and any data owner, with decisions recorded before a new data source or scenario is included.

Risk Impact Proposed control

Unapproved data High Use authorized/sanitized inputs

False positives Medium Analyst review and threshold tuning

Data drift Medium Periodic baseline review

Missing logs High Coverage inventory and exclusions

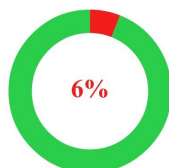
Score misuse High No automatic enforcement in prototype

3.5 Methodology Summary

The methodology establishes a repeatable sequence: define authorised data boundaries, normalise records, derive explainable features, calculate a proposed prioritisation signal, enrich an event with context and document the analyst decision. Each step can be inspected separately, which supports both academic evaluation and future controlled testing.

The method is deliberately conservative. It treats the algorithm as decision support and does not permit an automated score to alter a cloud ERP account, transaction or confi

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623	4749	1.86	29.14 (94%)

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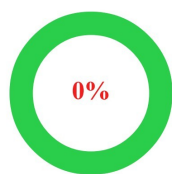
The literature also distinguishes between raw log retention and useful detection.

#2 3% Similar

<https://www.mdpi.com/1999-5903/16/5/167>

Proposed hybrid workflow Score plus human review Requires future pilot validation

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553	4216	0	25 (100%)

#1 100% Unique

CHAPTER 1

INTRODUCTION, SCOPE AND BACKGROUND

1. INTRODUCTION, SCOPE AND BACKGROUND

Cloud ERP platforms consolidate finance, procurement, inventory, human-resource and identity processes. This concentration improves visibility and process integration, but it also concentrates privileged access paths and audit responsibilities. A security event affecting a cloud ERP account can therefore have business, compliance and operational consequences beyond a single application.

This report presents a proposed AI-assisted cyber-defense prototype for real-time threat hunting in cloud ERP environments. The word prototype is important: the work documents a design, a structured analysis approach and illustrative scenarios. It does not claim a live enterprise deployment, independent production validation, measured detection accuracy or a completed incident-response integration.

1.1 Cloud ERP Security Context

Traditional monitoring commonly relies on fixed signatures, manually maintained rules and isolated alerts. These controls remain useful, but they can struggle when a sequence is unusual only in context: for example, a privileged action from a rare region immediately after a failed-login burst. Threat hunting adds a deliberate search for such patterns, using evidence from permitted audit trails.

The proposed system uses normalized audit events as its analytical unit. It aims to support human analysts by organizing

context and prioritizing potentially unusual activity. It is not designed to replace governance, access reviews, incident procedures or expert judgement.

1.2 Problem Definition

The central problem is not a lack of cloud telemetry; it is the difficulty of converting large, heterogeneous event streams into a manageable, explainable review queue. Cloud audit sources can record actors, actions, resources, times, locations, errors and request details, but these fields require preparation before they become useful for investigation.

Accordingly, the project asks how a prototype can combine data normalization, contextual feature engineering and anomaly-oriented scoring so that an analyst can investigate a small number of relevant events with traceable supporting context.

1.3 Aim and Objectives

Aim: to design an AI-assisted cyber-defense prototype for threat hunting in a cloud ERP context using authorized audit information. The project is framed as a design-and-analysis report rather than a claim of production implementation.

The objectives are to identify meaningful audit-log fields; define a normalisation and feature-engineering pipeline; describe an anomaly-oriented prioritisation approach; design analyst-facing triage information; and document limitations, controls and future validation requirements.

Objective Prototype output

Audit-log features Documented feature dictionary

Normalisation Traceable processing flow

Prioritisation Illustrative score rationale

Triage Analyst review design

1.4 Scope and Limitations

The scope is limited to approved, sanitized, synthetic or non-production audit data; conceptual integrations with cloud audit services; a prototype workflow for event review; and illustrative analytical figures. Any use of AWS CloudTrail, CloudWatch or an ERP source is described as a proposed or permitted-source design unless evidence is separately supplied.

Out of scope are autonomous containment in a production tenant, collection of personal data without approval, live response orchestration, claims of compliance certification, and accuracy benchmarking against a verified labelled enterprise dataset.

1.5 Significance of the Project

The project is significant because cloud ERP security is both a technical and business problem. The proposed workflow connects event telemetry with the language of business-critical actions, identities, access privileges and response ownership. It also highlights a practical principle: a score is only useful when an analyst can understand the event context behind it.

The report therefore treats explainability, data minimisation, auditability and human review as design requirements rather than optional features.